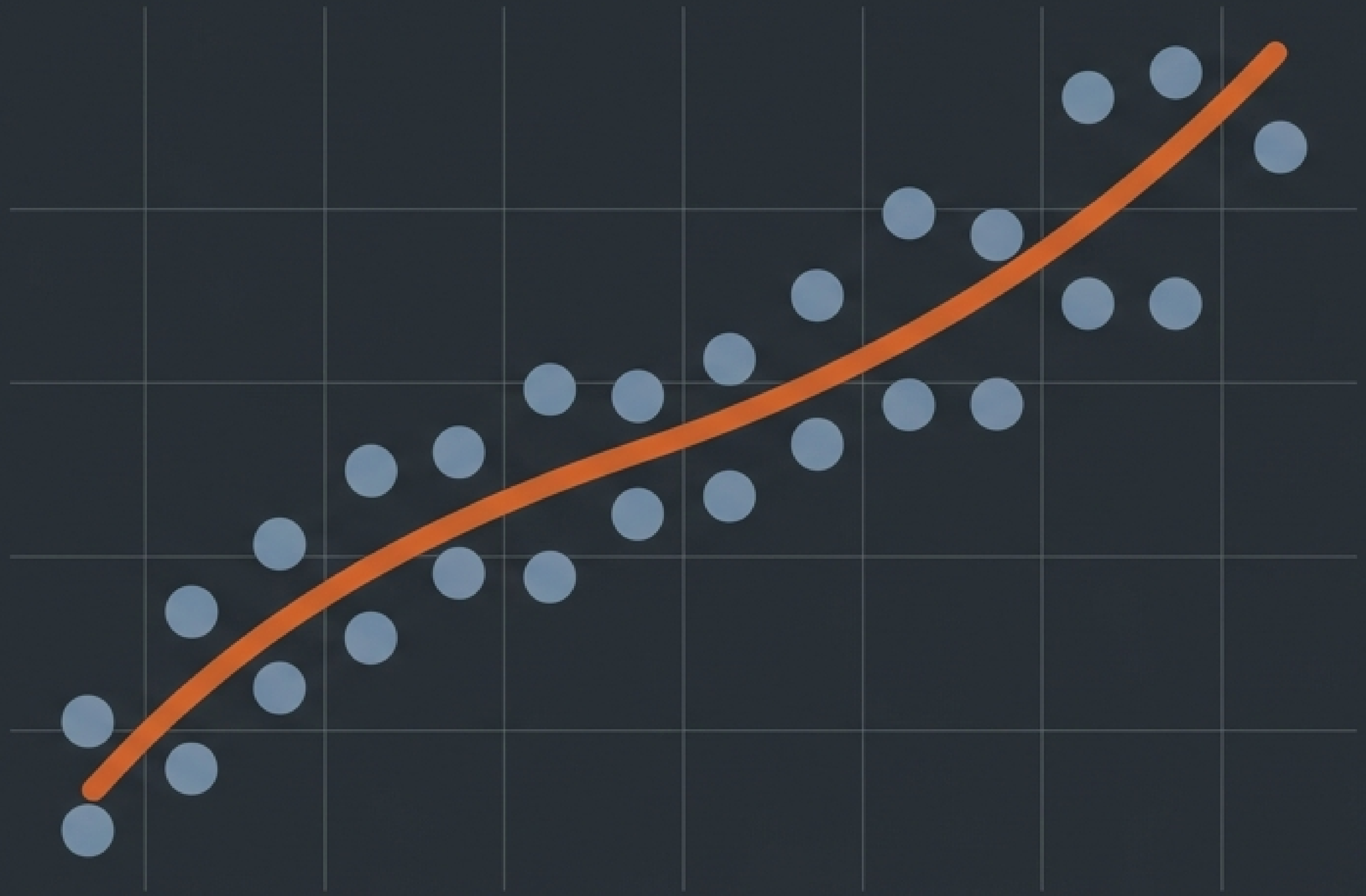
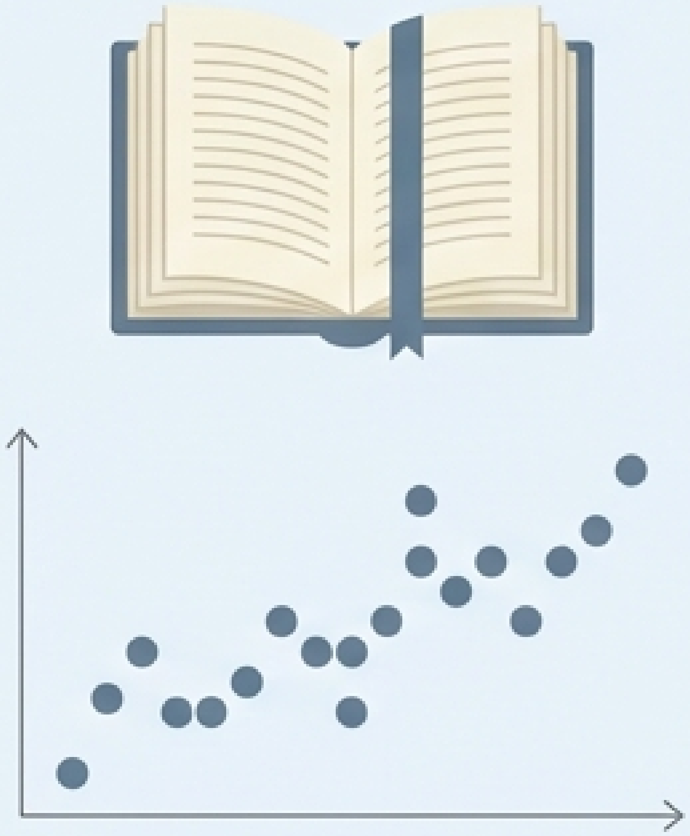
The Mechanics of Model Fit

A visual guide to Bias, Variance, and the reality of the Trade-off.

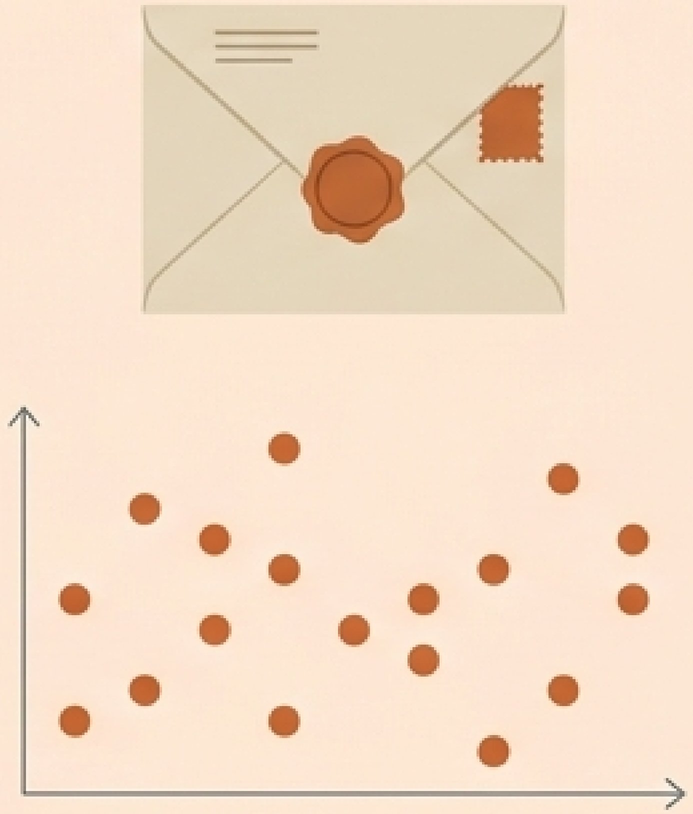


Models are trained on what we know, but evaluated on what they haven't seen.



Training Data

- The syllabus.
- Used to teach the model the underlying relationships.



Test Data (Unseen)

- The final exam.
- Used to prove the model can handle reality.

Evaluating a machine learning model is exactly like grading a student.

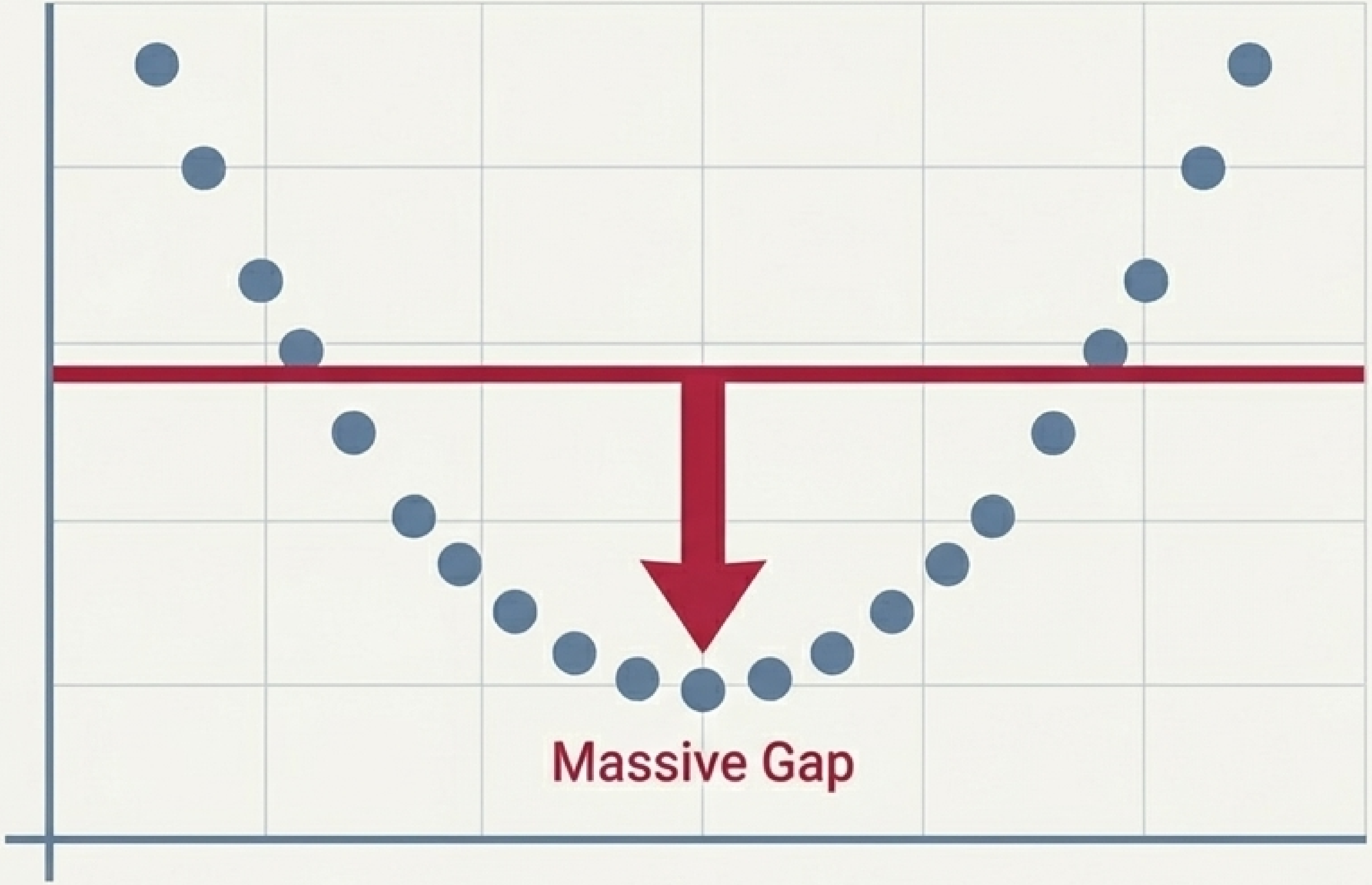
If a student fails the exam, we need to know why.

Did they refuse to study the textbook? Or did they just memorise the answers without understanding the logic?

Bias measures the inability to capture relationships in the Training Data.

High Bias means the model's foundational logic is too simplistic. It fails to learn the true patterns presented to it.

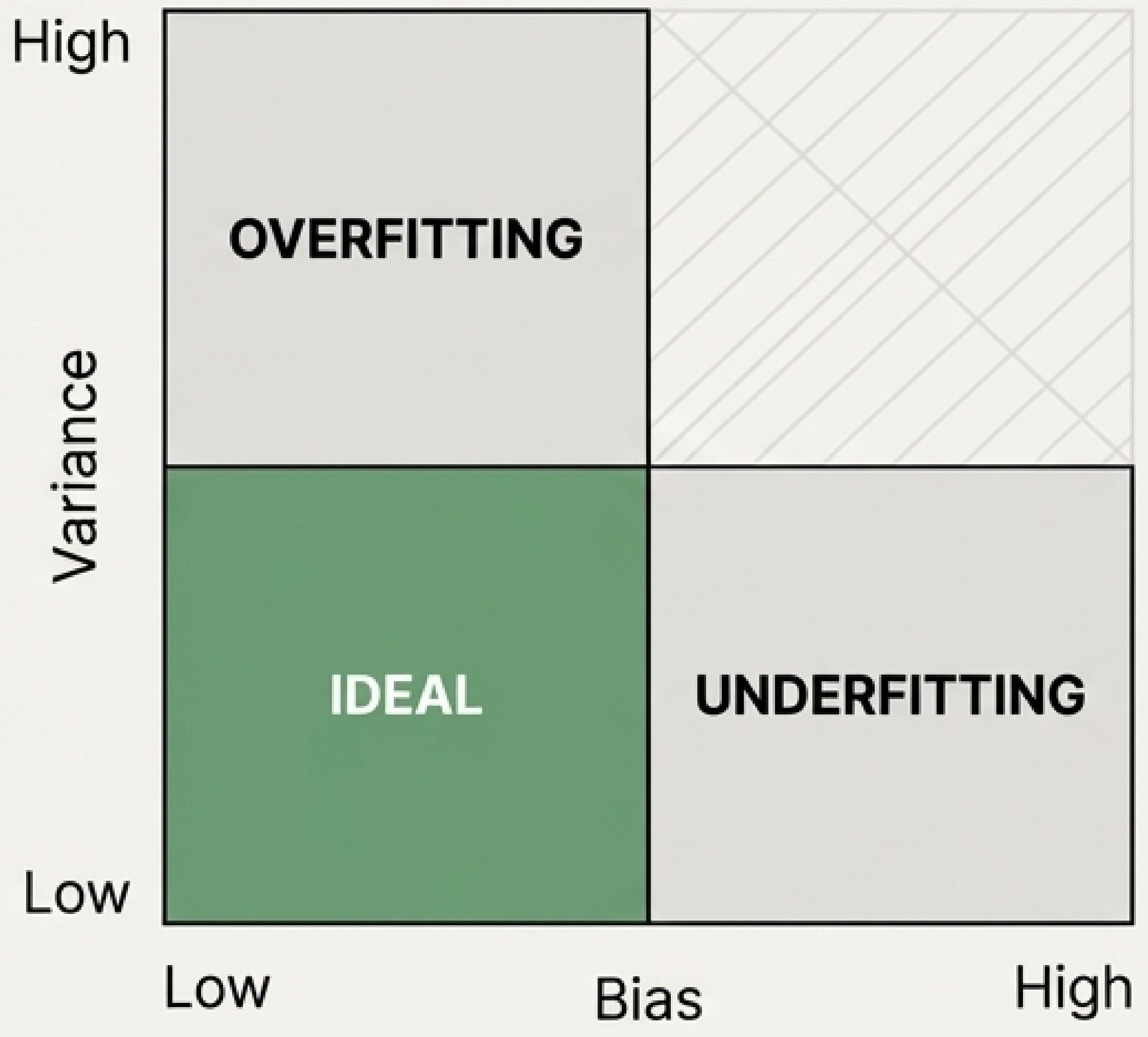
Like a student who couldn't grasp the basic syllabus.



Variance measures how wildly a model's performance fluctuates on Unseen Data.

High Variance means the model is hyper-sensitive. It captures every random fluctuation in the training set as if it were a strict rule.

The Model Fit Matrix maps how learning errors interact.

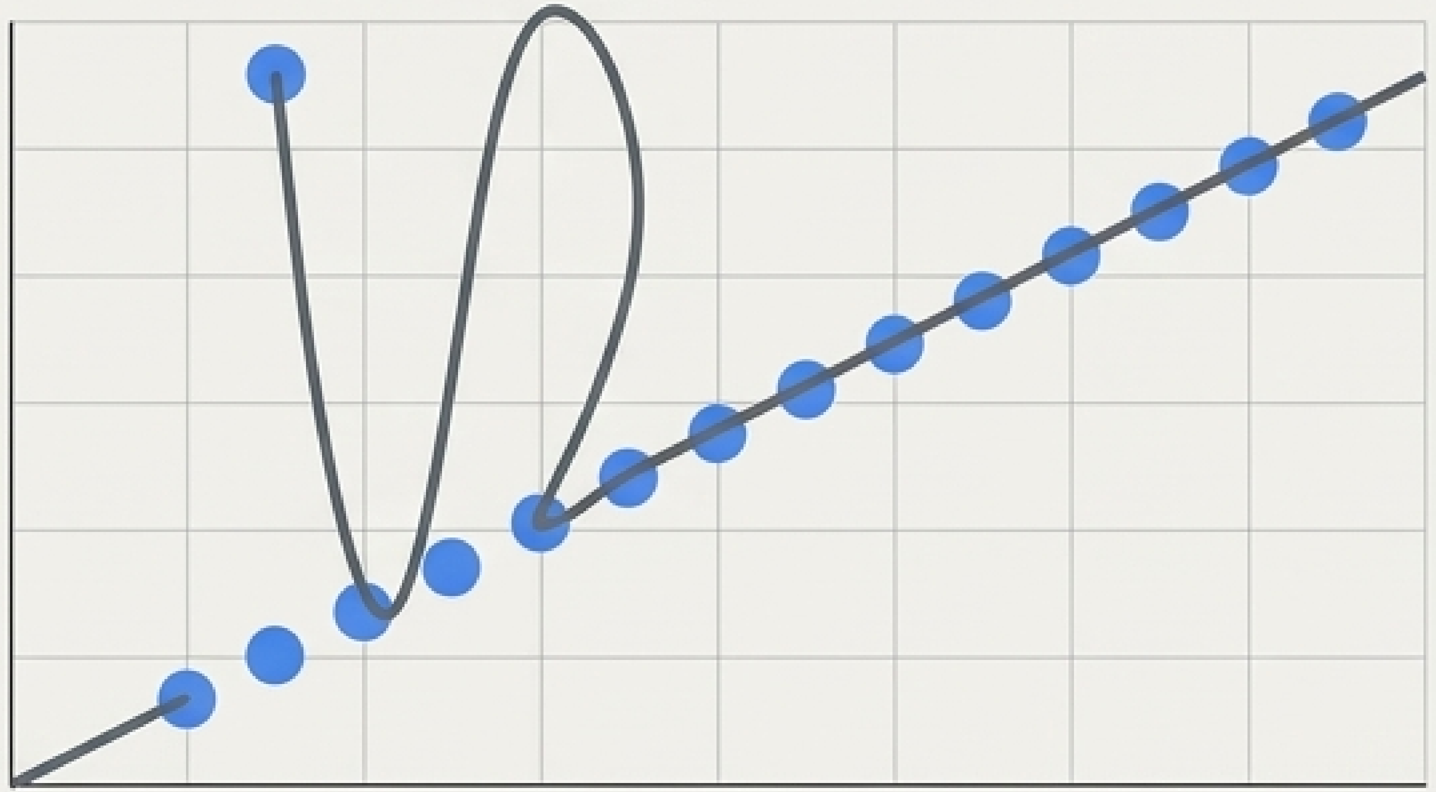


Understanding where a model sits on this matrix dictates how we fix it.

Lora headline

Overfitting occurs when models memorise the noise instead of the signal.

OVERFITTING
(Low Bias, High Variance)



Inter body

A Low Bias / High Variance model works too hard. By forcing its training error to zero, it completely loses the ability to generalise.

Give it a slightly different exam question, and it shatters.

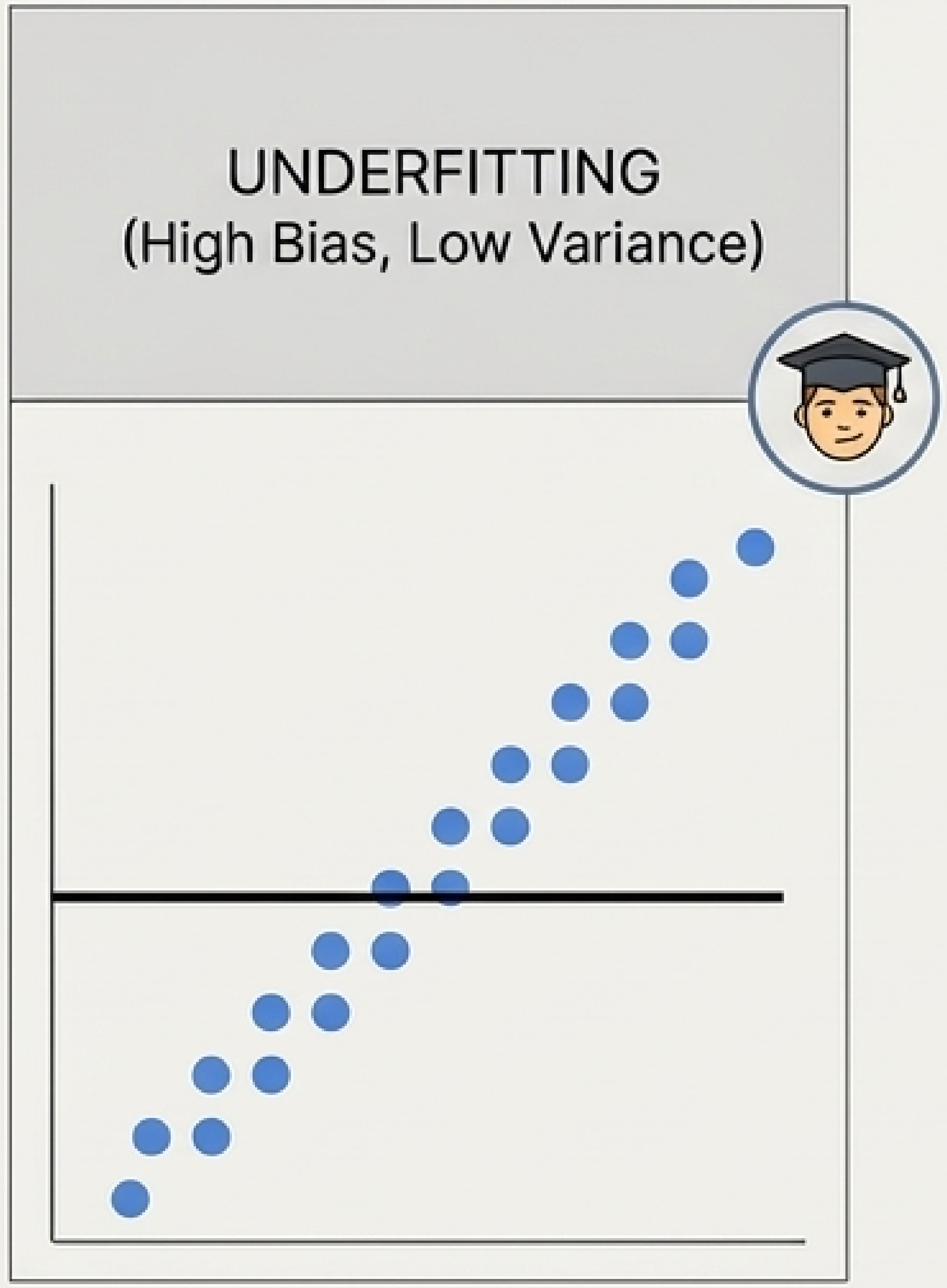
Lora

It memorises the data, but misses the concept.

Underfitting occurs when models are too rigid to learn anything useful.

A High Bias model is overly simplistic. It doesn't perform well on unseen data simply because it never bothered to capture the underlying relationships in the training data to begin with.

A failure to launch.



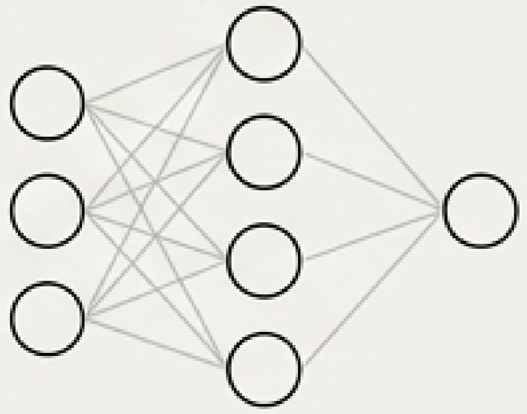
The root cause of fit issues is the complexity of the chosen model.

Bias Increases ⟵⟶ Variance Increases

Highly Simple Models

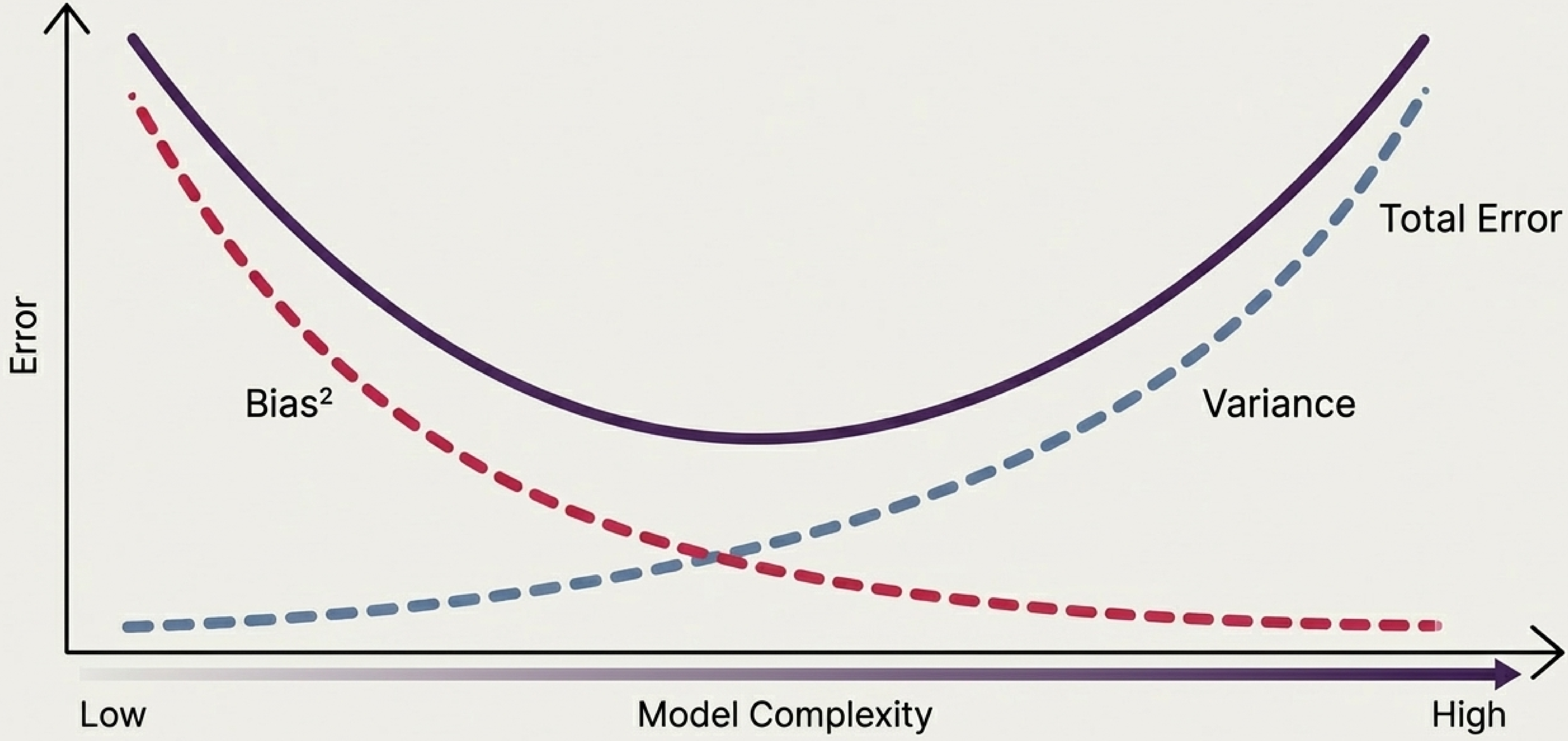
$$y = mx + c$$

Highly Complex Models

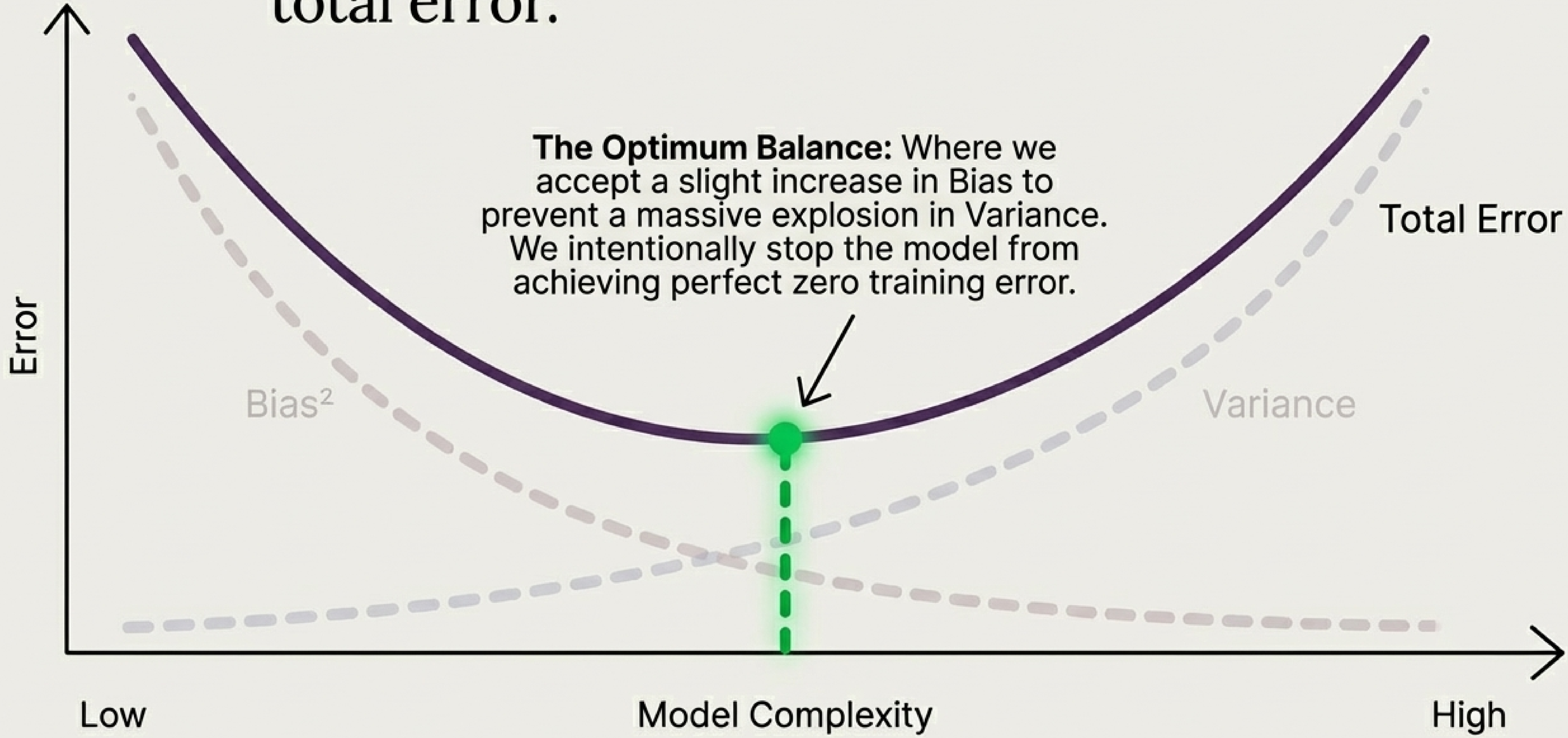


Simple models risk High Bias. Complex models invite High Variance.
You cannot adjust one without impacting the other.

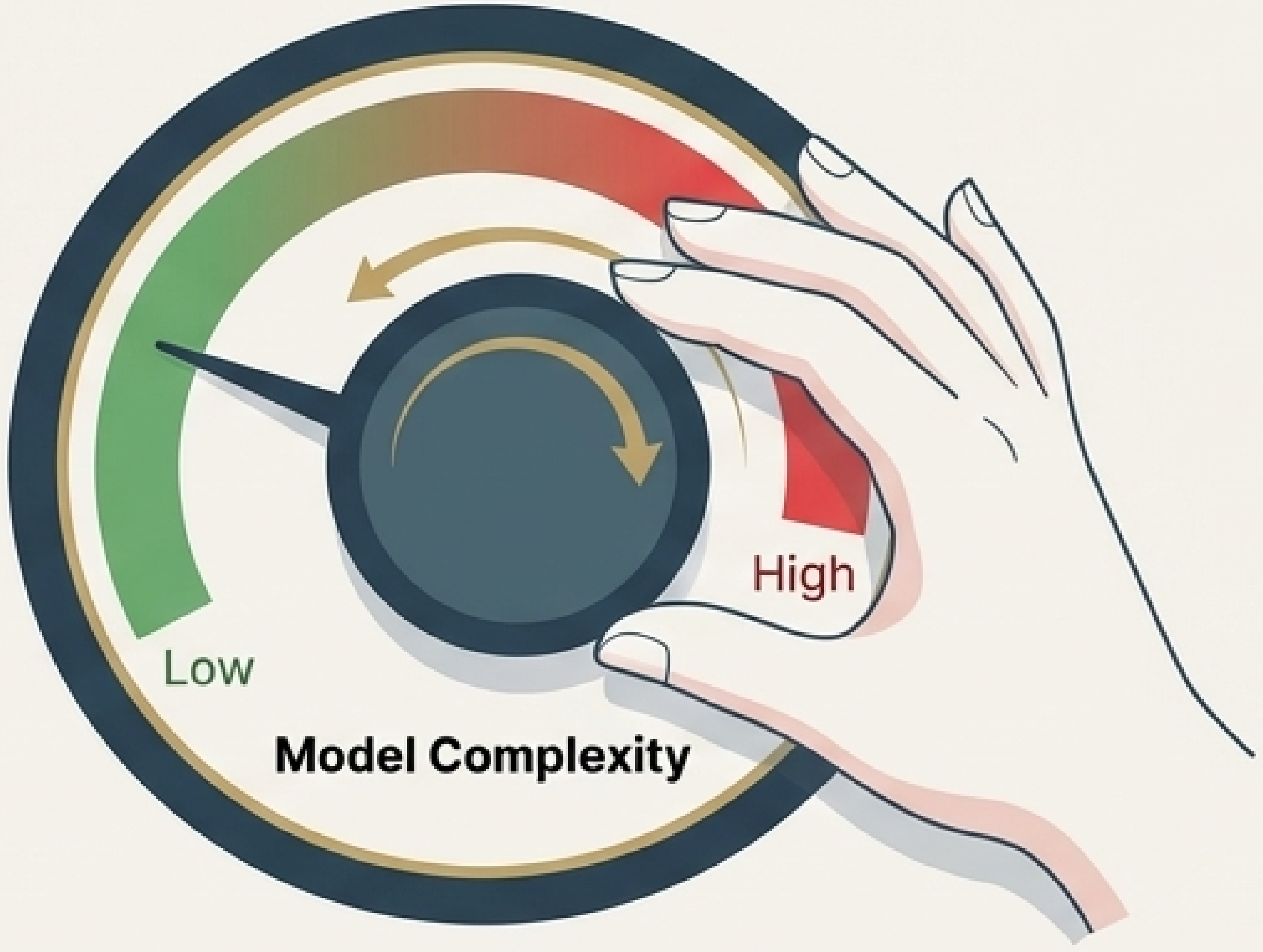
The Bias-Variance Trade-off curve.



The goal is always to find the sweet spot of minimum total error.



Regularisation is the mathematical brake pedal.



We don't just rely on luck to find the sweet spot. Techniques like Regularisation actively penalise a model for becoming too complex.

Lasso

Ridge

It forces the 'Memoriser' to forget the noise and focus only on the most critical patterns.